SARA BAKHTIARI

PhD Candidate in Mechanical Engineering | Cardiovascular Biomechanics | Medical Device R&D | Finite Element Modeling

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Eligible for U.S. internships through F-1 Curricular Practical Training (CPT), subject to university authorization.

PROFESSIONAL SUMMARY

PhD candidate in Mechanical Engineering at the University of Kentucky focused on cardiovascular biomechanics and computational modeling. Research spans myocardial constitutive behavior, ventricular mechanics, fiber architecture, efficiency, growth/remodeling, and fibrosis, with experience in finite-element analysis, validation, pressure-volume analysis, Python-based data analysis, and engineering R&D workflows.

EDUCATION

PhD, Mechanical Engineering (Direct PhD) - University of Kentucky | Spring 2024-Present
Candidate since Apr 2026 | GPA 3.91/4.00 | Advisor: Jonathan F. Wenk, PhD
B.Sc., Mechanical Engineering - Iran University of Science and Technology (IUST) | 2016-2021

RESEARCH EXPERIENCE

Graduate Research Assistant - Computational Cardiac Mechanics | Spring 2024-Present
University of Kentucky - Lexington, KY

- Develop and extend a coupled left-ventricular simulation framework integrating passive myocardial mechanics, active contraction, half-sarcomere kinetics, and cardiovascular circulation.
- Compare nonlinear anisotropic myocardial constitutive formulations to determine how material-law choice changes ventricular deformation, stress, pressure-volume behavior, and remodeling predictions.
- Develop material-validation workflows using biaxial, shear, and passive-inflation simulations to assess soft-tissue constitutive behavior before organ-level studies.
- Investigate myocardial fiber disarray and quantify changes in pump function, mechanical efficiency, ATP-related energetics, and baroreflex compensation.
- Develop an agent-based fibrosis model coupled with finite-element cardiac mechanics and analyze pressure-volume loops, stress/strain distributions, fiber architecture, and spatial outputs using Python and ParaView.

Machine Learning & Vision-Language Models for Electrospinning | 2026

- Apply vision-language models and regression methods to electrospinning video data for prediction of fiber morphology.
- Build repeatability and cross-validation workflows to evaluate robustness of model-derived features and predictions.

SELECTED RESEARCH PROJECTS

Passive Material & Organ-Level Validation: Use biaxial, shear, passive-inflation, stress/strain, and pressure-volume analyses to validate soft-tissue behavior before disease and remodeling studies.

Fiber Architecture & Pump Function: Quantify how myocardial fiber disarray alters pump function, mechanical efficiency, ATP-related energetics, and physiological compensation.

Fibrosis & Tissue Adaptation: Develop a coupled agent-based/finite-element model to study how spatial fibrosis changes myocardial mechanics and ventricular function.

PUBLICATIONS & RESEARCH OUTPUTS

Islam, R., Bakhtiari, S., Anugrah, G. G., Mehri, M., & Wenk, J. F. "Advancements in finite element modeling of myocardial infarction: a review." Biomechanics and Modeling in Mechanobiology, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0

Bakhtiari, S., Mehri, M., & Wenk, J. F. "Choice of Passive Myocardial Constitutive Law Drives Divergent Predictions of Left Ventricular Remodeling in Volume-Overload Conditions." Manuscript under review.

Bakhtiari, S., Mehri, M., & Wenk, J. F. "Effect of Myocardial Fiber Disarray on Left Ventricular Mechanical Efficiency and Baroreflex Compensation." Manuscript under review.

Selected presentation: "Choice of Passive Myocardial Constitutive Law Drives Differences in LV Mechanics and Growth Under Disease Conditions." Poster, BMES Annual Meeting, San Diego, CA, October 2025.

TEACHING EXPERIENCE

Teaching Assistant - Statics, University of Kentucky | Fall 2026

- Solved and explained undergraduate statics problems involving equilibrium equations, free-body diagrams, force/moment analysis, trusses, frames, and systematic engineering problem-solving methods.
- Supported classroom instruction, student questions, engagement, and class management; communicated technical concepts to students with different levels of understanding.
- Assisted with grading and examination supervision, evaluating engineering solution approaches and maintaining consistent exam procedures.

Teaching Assistant - Statics, Iran University of Science and Technology | 2021

- Supported undergraduate engineering-mechanics problem solving and helped students strengthen core statics concepts.

TECHNICAL SKILLS

Biomechanics & R&D: Cardiovascular mechanics, soft-tissue mechanics, anisotropic constitutive modeling, myocardial fiber architecture, fibrosis, growth/remodeling

Simulation & Validation: Nonlinear FEA, FEniCS/DOLFIN, material validation, biaxial/shear testing, passive inflation, pressure-volume analysis
Programming & Analysis: Python, MATLAB, NumPy, SciPy, Pandas, UFL
Engineering Software: COMSOL Multiphysics, SolidWorks, ParaView, Matplotlib; MPI/PETSc/SLURM HPC workflows